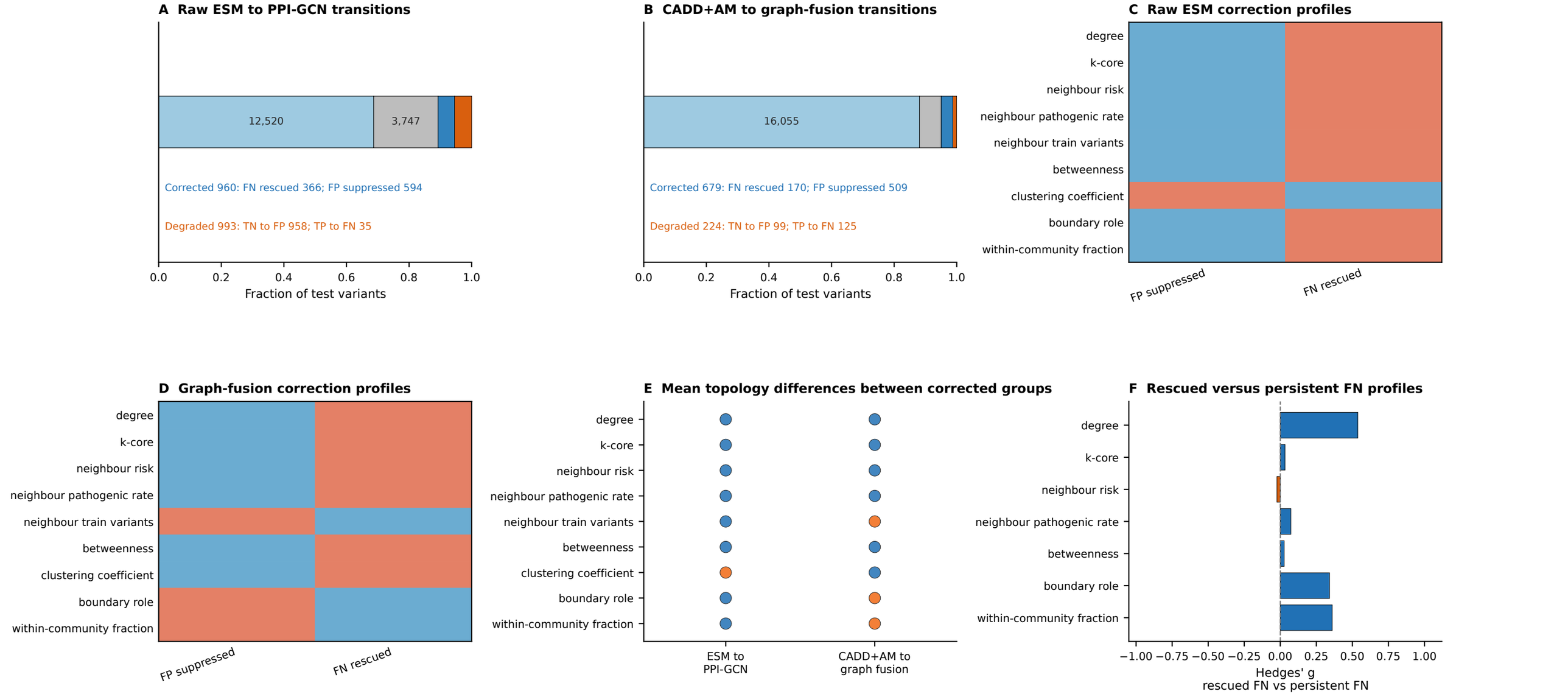Matched-ensemble prediction transitions and topology profiles